SentiMatrix LoRA Replication: Project Report

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1. Introduction and Objective

This project reproduces the main model of the paper *SentiMatrix* (LoRA fine-tuning of encoder transformers for sentiment analysis) on four of its datasets, using three random seeds per experiment on a Kaggle Tesla T4 GPU.

The goal was to check independently whether the paper's published LoRA results can be matched. Only the main LoRA model is reproduced. The paper's full fine-tuning, AdaLoRA and baseline comparisons are out of scope. Code, results and the README are in the GitHub repository *SentiMatrix-LoRA-Sentiment-Analysis*.

2. About the Paper

The paper shows that LoRA trains only small low-rank adapter matrices and the classifier head, cutting trainable parameters by up to 99.8% while staying close to full fine-tuning in accuracy.

- **Title:** SentiMatrix: Parameter-Efficient Fine-Tuning of Encoder-Based Transformers for Multidimensional Sentiment Analysis
- **Authors:** Md. Easin Arafat, Muhammad Usman Akmal, Ali S. Abosinnee, Tamás Orosz
- **Venue:** Machine Learning (2026) 115:229, DOI 10.1007/s10994-026-07161-4
- **Authors' code:** github.com/ELTE-DSED/senti-matrix
- **Scope of the paper:** seven datasets and five encoder models, comparing LoRA, AdaLoRA and full fine-tuning

LoRA freezes the pretrained weights and adds a trainable low-rank update (rank r, scaling alpha) to selected attention layers. Only these adapters and the classification head are trained.

3. Datasets

Four datasets from the authors' repository were used, each split 80/10/10 into train, validation and test sets.

Dataset	Task	Classes	Rows used	Train / Val / Test	Model
SST-2	Sentiment	2	9,613	7,690 / 961 / 962	DistilBERT, RoBERTa-base
CARER Emotion	Emotion detection	6	25,000 (subset)	20,000 / 2,500 / 2,500	DistilBERT
Amazon E-Commerce	Review rating	5	22,641	18,112 / 2,264 / 2,265	BERT-base-multilingual
SemEval-2014 ABSA	Aspect-based sentiment	3	5,914	4,731 / 591 / 592	DeBERTa-v3-base

The Emotion file in the authors' repository is a 25,000-row subset, while the paper reports 75,000 rows.

4. Methodology

Each experiment follows the paper's per-dataset LoRA settings (its Table 3b) in five steps.

1. **Data split.** 80% train, 10% validation, 10% test, stratified with `random_state=42`. This gives the same test set as the authors' code.
2. **LoRA setup.** Adapters are added to the attention projections with the `peft` library, and the classifier head is fully trained. Rank, alpha, dropout, learning rate, batch size, epochs and weight decay follow the paper for each dataset.
3. **Training.** fp16 mixed precision, cosine learning-rate schedule, warmup of 10% (3% for Emotion), early stopping on validation loss (patience 3, or 1 for Emotion). Each experiment is run with seeds 42, 43 and 44.
4. **Evaluation.** On the held-out test set: accuracy, precision, recall and F1 (macro-averaged, weighted for ABSA), similarity, confidence, ECE, Brier score, trainable parameters, training time and GPU memory.
5. **Reporting.** Mean ± standard deviation over the three seeds, compared with the paper's published values.

Main LoRA settings:

Setting	SST-2	Emotion	E-Commerce	ABSA
Rank / alpha	6 / 12	8 / 16	8 / 16	6 / 12
LoRA dropout	0.2	0.1	0.05	0.1

Setting	SST-2	Emotion	E-Commerce	ABSA
Batch size (train / eval)	8 / 16	32 / 64	8 / 16	8 / 16
Max epochs	10	5	3	10
Weight decay	0.05	0.01	0.01	0.2

Metric definitions used: Similarity is the cosine between the predicted probability vector and the one-hot label. Confidence is the mean maximum softmax probability. ECE uses 10 equal-width bins. Brier score is the mean squared error between probabilities and one-hot labels, summed over classes.

5. Implementation

Everything runs in one Kaggle notebook (Tesla T4 GPU, Internet on). It clones the authors' repository for the datasets, trains LoRA on all four datasets with three seeds (about 75 minutes), and writes tables and figures to an output folder that is zipped for download.

Problems met on the way, and the fixes:

Problem	Fix
`peft` failed to import because Kaggle's `torchao` was too old	Uninstall `torchao` and restart the session
`warmup_ratio` rejected by the newest `transformers`	Small compatibility helper for the training arguments
Two GPUs silently doubled the batch size (half the training steps)	Set `CUDA_VISIBLE_DEVICES=0`
DeBERTa-v3 loaded in fp16 and crashed with "unscale FP16 gradients"	Cast the model to fp32 right after loading
Pinned old library versions could not be installed on Python 3.13	Use the libraries preinstalled on Kaggle
My first metrics compared binary F1 and a true-class confidence against the paper	Switched to macro-F1 and mean max-softmax confidence, which match the paper's tables

After the fixes, all four datasets with three seeds ran without errors.

6. Results

Accuracy and F1 are within about 1 to 2 points of the paper on every dataset, and the trainable-parameter counts match the paper exactly. Values are test-set percentages, mean ± std over 3 seeds (the paper's ABSA row is a single run).

Dataset / model	Accuracy: paper	Accuracy: ours	F1: paper	F1: ours	Trainable parameters: paper → ours
SST-2, DistilBERT	89.02 ± 0.95	88.98 ± 0.31	89.22 ± 1.18	88.96 ± 0.30	758K → 758,018
SST-2, RoBERTa-base	93.28 ± 0.91	94.07 ± 0.63	93.51 ± 0.91	94.06 ± 0.63	1.20M → 1,200,386
Emotion, DistilBERT	93.22 ± 0.03	92.85 ± 0.18	90.20 ± 0.03	89.66 ± 0.16	742.66K → 742,662
E-Commerce 5-class, mBERT	67.63 ± 0.26	67.26 ± 0.16	47.37 ± 1.48	49.31 ± 0.87	298.76K → 298,757
ABSA, DeBERTa-v3	80.74	79.73 ± 0.45	79.84	78.66 ± 0.78	334.08K → 334,083

Calibration and similarity results (ours, mean ± std):

Dataset / model	Similarity (%)	Confidence (%)	ECE (%)	Brier
SST-2, DistilBERT	90.82 ± 0.09	93.92 ± 0.78	5.07 ± 0.78	0.168 ± 0.002
SST-2, RoBERTa-base	94.76 ± 0.56	97.99 ± 0.26	3.97 ± 0.29	0.099 ± 0.011
Emotion, DistilBERT	94.84 ± 0.23	94.89 ± 0.04	2.07 ± 0.26	0.089 ± 0.004
E-Commerce 5-class, mBERT	73.92 ± 0.31	69.62 ± 2.13	3.70 ± 1.06	0.431 ± 0.004
ABSA, DeBERTa-v3	82.51 ± 0.17	84.31 ± 2.96	6.18 ± 2.18	0.309 ± 0.005

The confusion matrices, training curves and ours-versus-paper plots for each dataset are in the repository under `Results/Graphs`.

7. Observations and Limitations

LoRA reaches near-paper accuracy while training well under 1.5 million parameters per model.

Observations:

- Parameter counts match the paper exactly for all five models, which confirms the LoRA configuration is the same.
- RoBERTa on SST-2 scored higher than the paper (94.07 vs 93.28). DistilBERT matched it (88.98 vs 89.02).
- E-Commerce 5-class has decent accuracy but low macro-F1, because the minority rating classes are hard. The paper shows the same pattern.
- Seed-to-seed variance on SST-2 is smaller in our runs than in the paper's.

Limitations:

- **Emotion** used the 25,000-row subset from the authors' repository, not the paper's 75,000 rows. Our similarity (94.84) is 2.4 points above the paper's, and this gap is unexplained.
- **ABSA** results in the paper are a single run, so the comparison is less strict. Our ECE and Brier are somewhat worse.
- **ECE and Brier** follow my own implementation, since the paper does not fully specify them.
- **Training time and GPU memory** depend on hardware and are not directly comparable.
- Twitter, IMDb, Yelp, E-Commerce 3-class, full fine-tuning and AdaLoRA are not reproduced.

8. Conclusion and Reference

The paper's LoRA results were reproduced on four datasets (five model and dataset pairs): trainable-parameter counts match exactly, and accuracy and F1 land within about 1 to 2 points of the published values. The remaining differences come from the smaller Emotion subset, the single-run ABSA numbers in the paper, unspecified calibration-metric details and different hardware.

Reference: Arafat, M. E., Akmal, M. U., Abosinnee, A. S., Orosz, T. (2026). SentiMatrix: Parameter-Efficient Fine-Tuning of Encoder-Based Transformers for Multidimensional Sentiment Analysis. *Machine Learning*, 115, 229. https://doi.org/10.1007/s10994-026-07161-4